

St. Francis Institute of Technology, Mumbai-400 103
Department Of Information Technology

A.Y. 2025-2026
Class: SE-ITA/B, Semester: IV

Subject: **DevOps Lab**

Experiment – 8: To execute test cases on a browser using Selenium IDE and set up and run Selenium tests in Jenkins using Maven.

1. **Aim:** To execute test cases on a browser using Selenium IDE and set up and run Selenium tests in Jenkins using Maven.
2. **Objectives:** Aim of this experiment is that, the students will learn:
 - Selenium and how to automate your test cases for testing web elements
3. **Outcomes:** After study of this experiment, the students will learn following:
 - Introduction to Selenium
 - Installing Katalon (Selenium Tests generator)
 - Creating Test Cases in Selenium WebDriver
 - Run Selenium Tests in Jenkins Using Maven
4. **Prerequisite:** Knowledge of Software Engineering concept of testing and test cases.
5. **Requirements:** Jenkins, JDK, Eclipse IDE, Chrome browser, Personal Computer, Windows operating system, Internet Connection, Microsoft Word.
6. **Pre-Experiment Exercise:**
Brief Theory: Refer shared material
7. **Laboratory Exercise**
 - A. **Procedure:**
 - a. **Answer the following:**
 - Explain Selenium suite?
 - What are the limitations of Selenium IDE?
 - b. **Execute following (Refer the shared material) and attach screenshots:**
 - Create and run a test case on Chrome/Firefox browser with Katalon Recorder(selenium tests generator) IDE addon
 - Create a Maven Project in Jenkins and run selenium tests using selenium Grid
8. **Post-Experiments Exercise**
 - A. **Extended Theory:**
Nil
 - B. **Questions:**
 - What are Locators? Explain its types.
 - What is the benefit of using Selenium Grid with Jenkins?
 - C. **Conclusion:**
 - Write what was performed in the experiment.
 - Write the significance of the topic studied in the experiment.
 - D. **References:**
 - <https://jenkins.io/doc/>
 - <https://www.slideshare.net/abediaz/introduction-to-jenkins>
 - <https://q-automations.com/2019/09/26/selenium-grid-with-jenkins/>

7. Laboratory Exercise

A. Procedure:

a. Answer the following:

1. Explain Selenium Suite

The **Selenium Suite** is a collection of tools used for automating web applications for testing purposes. It is widely used for functional and regression testing of web applications across different browsers and platforms. The suite consists of the following components:

1. **Selenium IDE (Integrated Development Environment)**
 - It is a browser extension (mainly for Chrome/Firefox) used for recording and playback of test cases.
 - It is beginner-friendly and does not require programming knowledge.
 - Useful for quick prototyping of test scripts.
2. **Selenium WebDriver**
 - It is the most important component of the Selenium suite.
 - It allows writing test scripts in multiple programming languages like Java, Python, C#, etc.
 - It directly interacts with the browser and provides better control and performance compared to Selenium IDE.
3. **Selenium RC (Remote Control) (Deprecated)**
 - Earlier used to run tests in different browsers using a server.
 - Now replaced by WebDriver due to limitations like slower execution and complex architecture.
4. **Selenium Grid**
 - Used to run tests on multiple machines and browsers simultaneously.
 - Helps in parallel execution, reducing testing time.
 - Useful for cross-browser and distributed testing.

Overall, the Selenium Suite provides flexibility, scalability, and support for multiple environments, making it a powerful tool for web application testing.

2. Limitations of Selenium IDE

Although Selenium IDE is easy to use, it has several limitations:

1. **Limited to Browser Automation Only**
 - Selenium IDE can only automate web applications.
 - It cannot test desktop applications or mobile apps.
2. **No Advanced Programming Logic**
 - It does not support complex logic like loops, conditions, or data-driven testing effectively.
 - This limits its use for large-scale testing.
3. **Lack of Reusability and Maintainability**
 - Test scripts created in Selenium IDE are not easily reusable.
 - Maintenance becomes difficult as test cases grow.

4. **Limited Browser Support**
 - Mainly supports Chrome and Firefox.
 - Not suitable for extensive cross-browser testing.
5. **No Integration with Testing Frameworks**
 - Cannot integrate with frameworks like TestNG or JUnit directly.
 - This restricts structured test execution and reporting.
6. **Weak Error Handling and Debugging**
 - Provides limited debugging capabilities.
 - Difficult to handle dynamic elements or unexpected failures.
7. **Not Suitable for Large Projects**
 - Works well for small or simple test cases.
 - Not recommended for enterprise-level or complex automation.

In summary, Selenium IDE is useful for beginners and quick testing, but for advanced and scalable automation, Selenium WebDriver is preferred.

Jenkins Manage Jenkins



Jenkins

+ New Item

Build History

Build Queue (1)

Master Slave1

Build Executor Status

Built-In Node

0/2

DevOpsexp6

(offline)

Leonslv

(offline)

jaden_slv

(offline)

slave1

(offline)

slave69

(offline)

slave786

(offline)

AllChrisJadenKimayaLeon+

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	ant project	1 mo 2 days #1	N/A	8.9 sec ▶
✓	☀	AntProject	1 mo 2 days sherwin	N/A	5.5 sec ▶
✓	☀	DemoScript ▾	18 days #3	N/A	0.71 sec ▶
✓	☀	EXP7A	17 days #3	N/A	0.26 sec ▶
✓	☀	EXP7Ab	18 days #6	N/A	5.1 sec ▶
✓	☀	freestyle fresh	1 mo 2 days #1	N/A	0.24 sec ▶
✓	☀	GitFork	18 days #1	N/A	3.9 sec ▶
✓	☀	HelloLeon	1 mo 4 days #3	N/A	74 ms ▶
✓	☁	HR_Java	1 mo 5 days #6	1 mo 5 days #2	0.69 sec ▶
✓	☁	HR_Python	1 mo 5 days #9	1 mo 5 days #5	0.11 sec ▶

Click on plugins



Jenkins / Manage Jenkins

Q Search settings

7

Manage Jenkins

The following installed plugins are deprecated:
[Python Plugin](#)
In general, this means that these plugins are either obsolete, no longer being developed, or may no longer work. See the linked web pages for further information about the cause for the deprecation, and suggestions on how to proceed.

New version of Jenkins (2.554) is available for [download](#) ([changelog](#)).

Or Upgrade Automatically

Building on the built-in node can be a security issue. You should set the number of executors on the built-in node to 0. See [the documentation](#).

ManageDismiss

Warnings have been published for the following currently installed components:

Go to plugin manager

Configure which of these warnings are shown

Jenkins 2.533 core and libraries:
[Multiple security vulnerabilities in Jenkins 2.550 and earlier, LTS 2.541.1 and earlier](#)
[Multiple security vulnerabilities in Jenkins 2.540 and earlier, LTS 2.528.2 and earlier](#)
Fixes for all of these issues are available. Update Jenkins now.

Selenium Plugin 3.141.59:
[Complete lack of CSRF protection can lead to OS command injection \(no fix available\)](#)
No fixes for these issues are available. It is recommended that you review the security advisory and apply mitigations if possible, or uninstall this plugin.

Install plugin : Selenium and Selenium HTML Report

Jenkins

Manage Jenkins

Plugins

Search

Sele

Updates

Available plugins

Installed plugins

Advanced settings

Name

Health

Enabled

Selenium HTML report 1.1

This is an Jenkins plugin to visualize the results of selenium tests
Report an issue with this plugin

45

Selenium Plugin 3.141.59

Report an issue with this plugin

Warning: The currently installed plugin version may not be safe to use. Please review the following security notices:

- Complete lack of CSRF protection can lead to OS command injection

34

This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.

Jenkins More actions selenium grid

Jenkins

+ New Item

Build History

Build Queue (1)

Master Slave1

Build Executor Status

Built-In Node 0/2

DevOpsxp6 (offline)

Leonslv (offline)

jaden_slv (offline)

slave1 (offline)

slave69 (offline)

slave786 (offline)

All Chris Jaden Kimaya Leon +

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	ant project	1 mo 1 day #1	N/A	8.9 sec
✓	☀	AntProject	1 mo 2 days sherwin	N/A	5.5 sec
✓	☀	DemoScript	18 days #3	N/A	0.71 sec
✓	☀	EXP7A	17 days #3	N/A	0.26 sec
✓	☀	EXP7Ab	18 days #6	N/A	5.1 sec
✓	☀	freestyle fresh	1 mo 2 days #1	N/A	0.24 sec
✓	☀	GitFork	18 days #1	N/A	3.9 sec
✓	☀	HelloLeon	1 mo 3 days #3	N/A	74 ms

Search

Selenium Grid

option

 Plugin home Configurations Nodes matching configurations New configuration

New configuration

Name of the configuration

Arthak

Matcher

- ☒ Match all nodes ?
- ☐ Match no nodes ?
- ☐ Match nodes from a label expression ?

Type

- ☐ Custom RC node configuration ?
- ☒ Custom web driver node configuration ?

Preferred port ?

4445

Max node sessions

5

Save configuration

 Plugin home Configurations Nodes matching configurations New configuration

Type	Delete	Name ↓	Matching type	Description summary
		jaden	Match all nodes	1 instances of Internet Explorer (version : Not specified) 5 instances of Firefox (version : Not specified) 5 instances of Chrome (version : Not specified)
		Kimaya	Match all nodes	1 instances of Internet Explorer (version : Not specified) 5 instances of Firefox (version : Not specified) 5 instances of Chrome (version : Not specified)
		Leon	Match all nodes	1 instances of Internet Explorer (version : Not specified) 5 instances of Firefox (version : Not specified) 5 instances of Chrome (version : Not specified)
		Meet	Match all nodes	1 instances of Internet Explorer (version : Not specified) 5 instances of Firefox (version : Not specified) 5 instances of Chrome (version : Not specified)

 chrome web store

Discover

Extensions

Themes



Katalon Recorder (Selenium tests generator)

Add to Chrome

 **Featured**

4.2★ (267 ratings)



 Share

Extension

Workflow & Planning

100,000 users

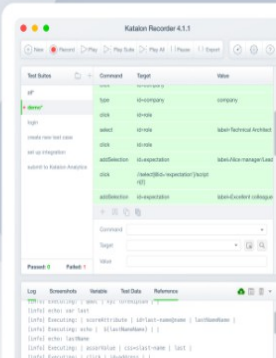


 **Katalon Recorder**

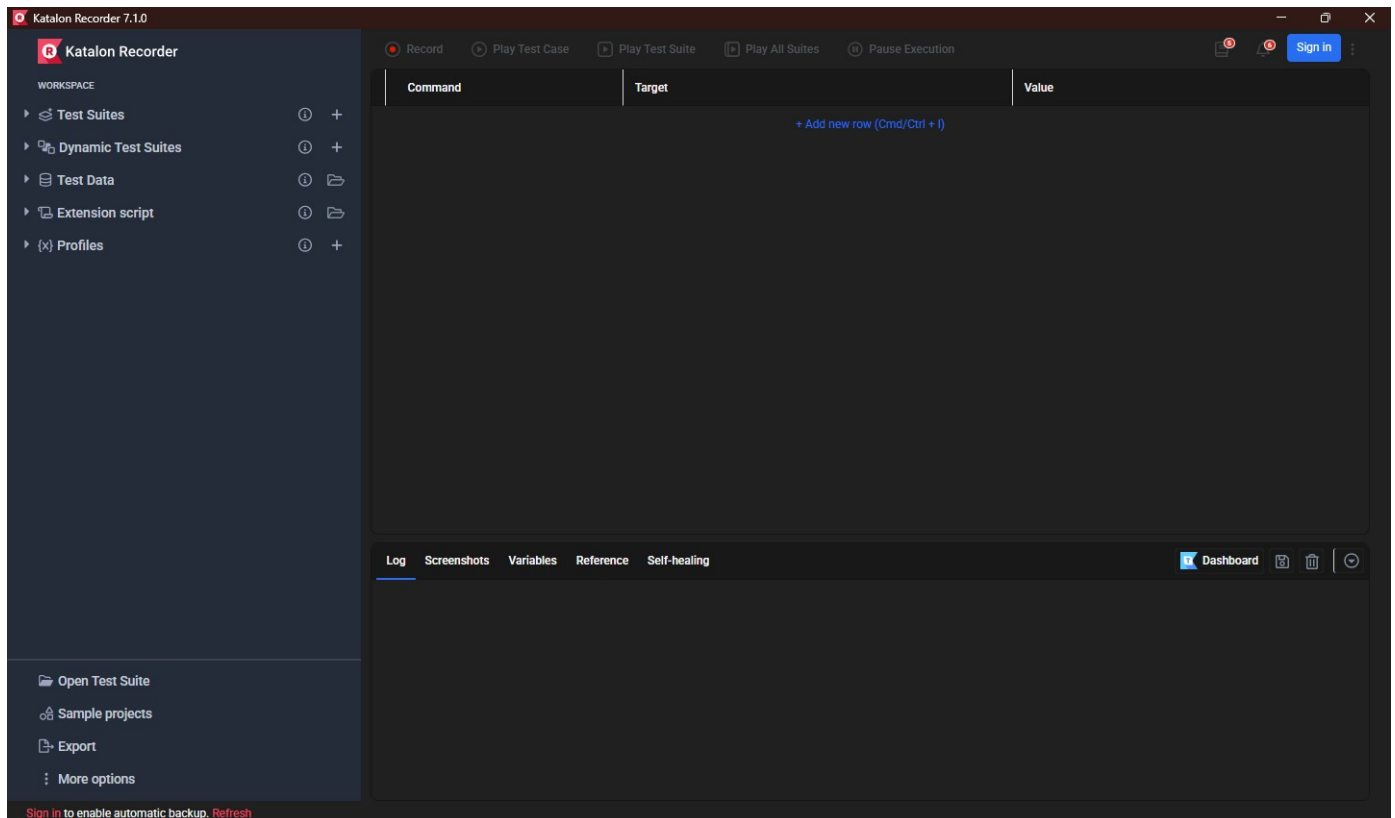
Powerful recorder to capture & edit automation scripts.

Manage and Run test suite easily.

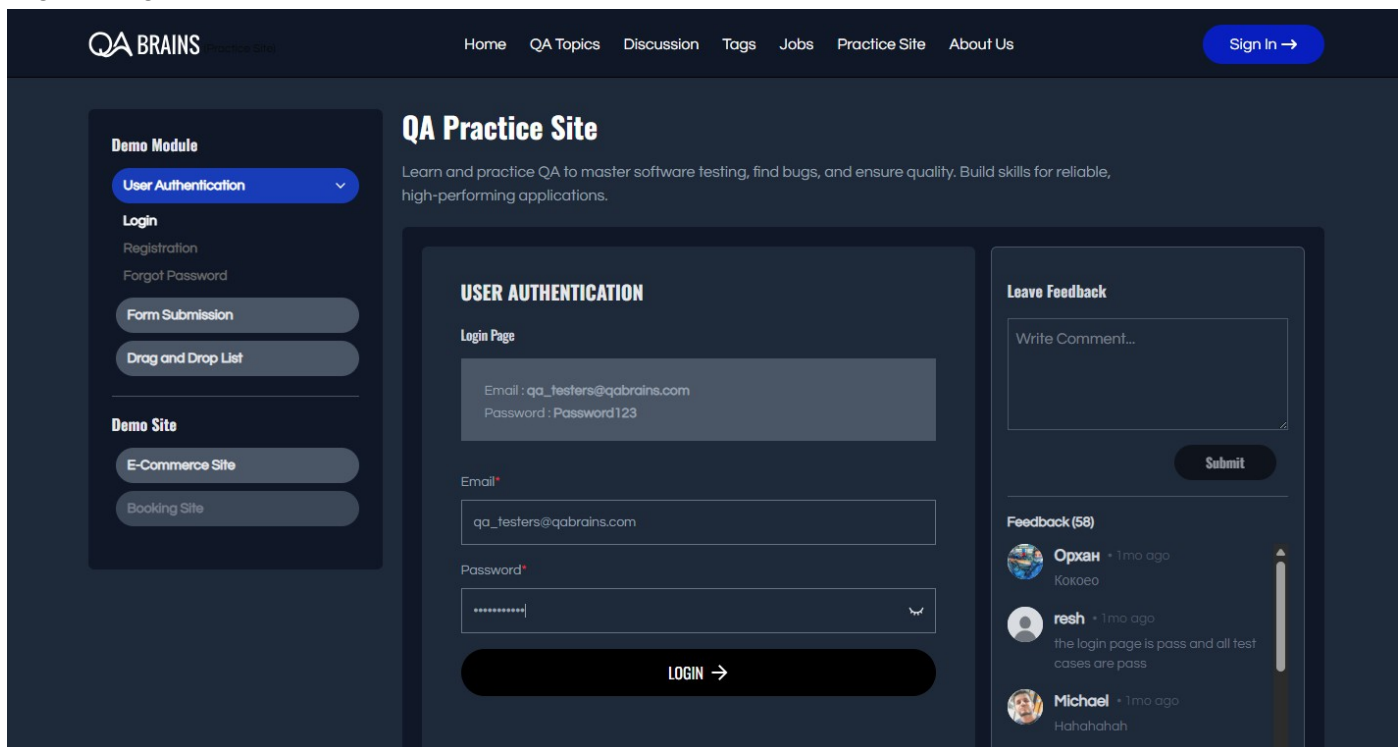
Get automation tools at katalon.com



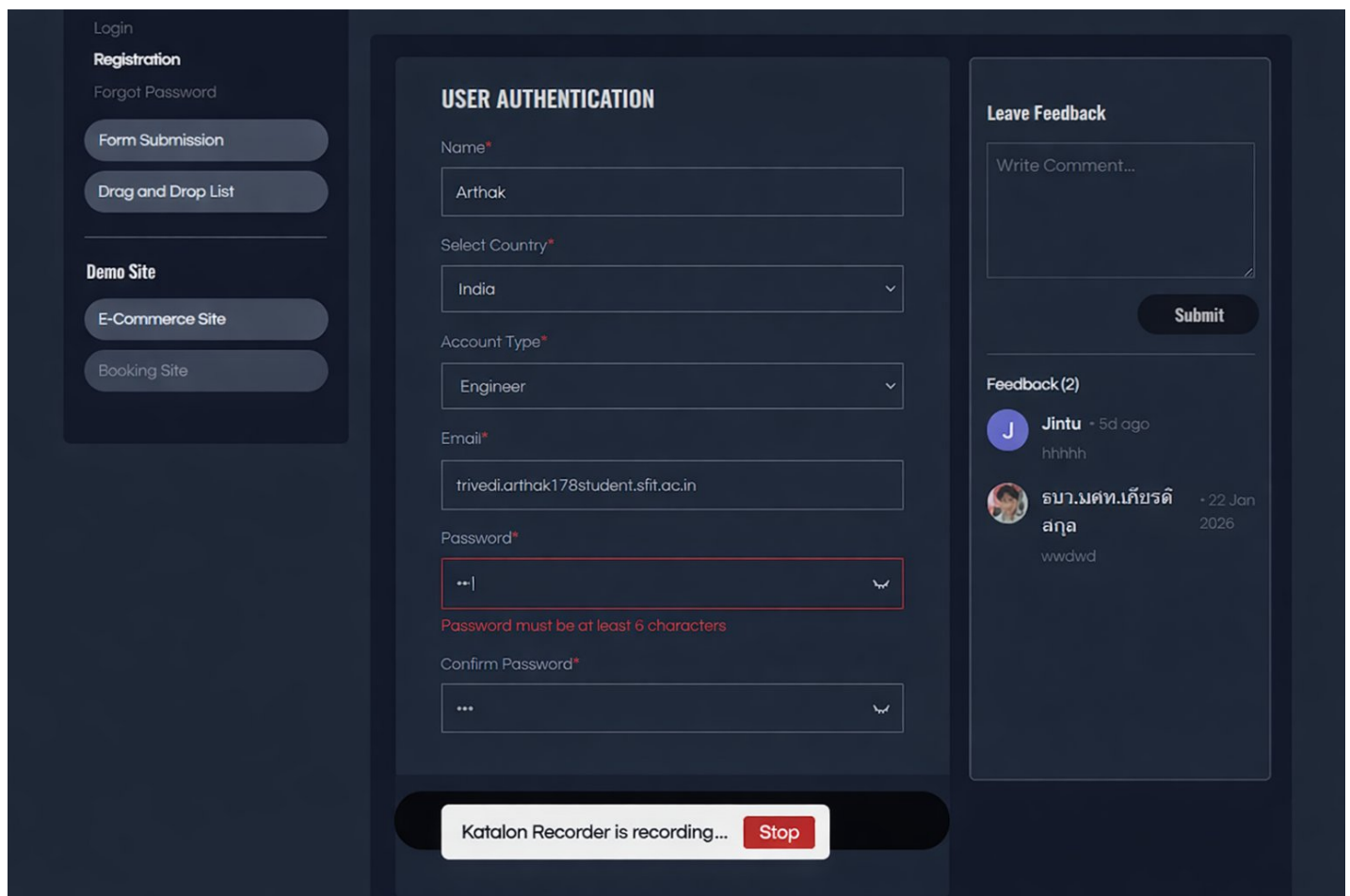
Katalon Recorder



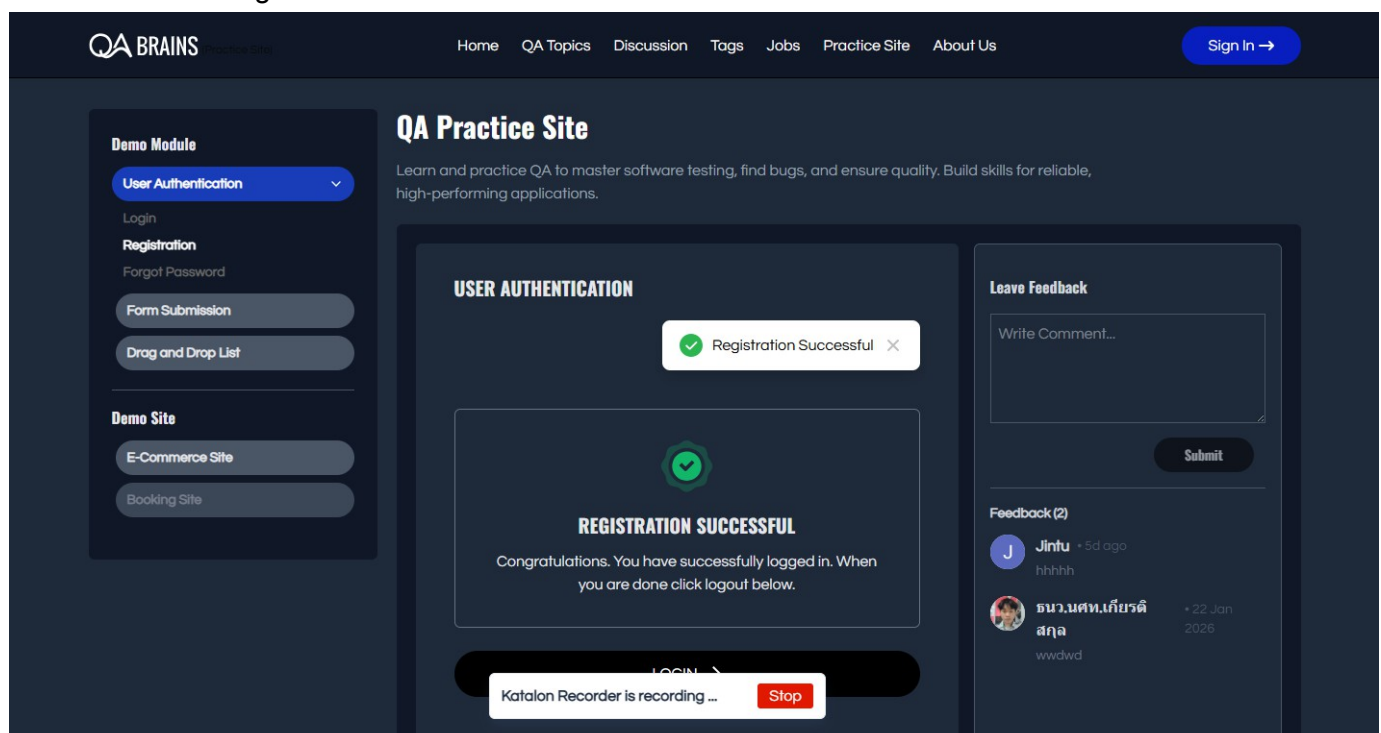
Go to website QA Brain Practice site
Login with given username and password



Go to Katalon and press record button
Click on Registration and fill all details

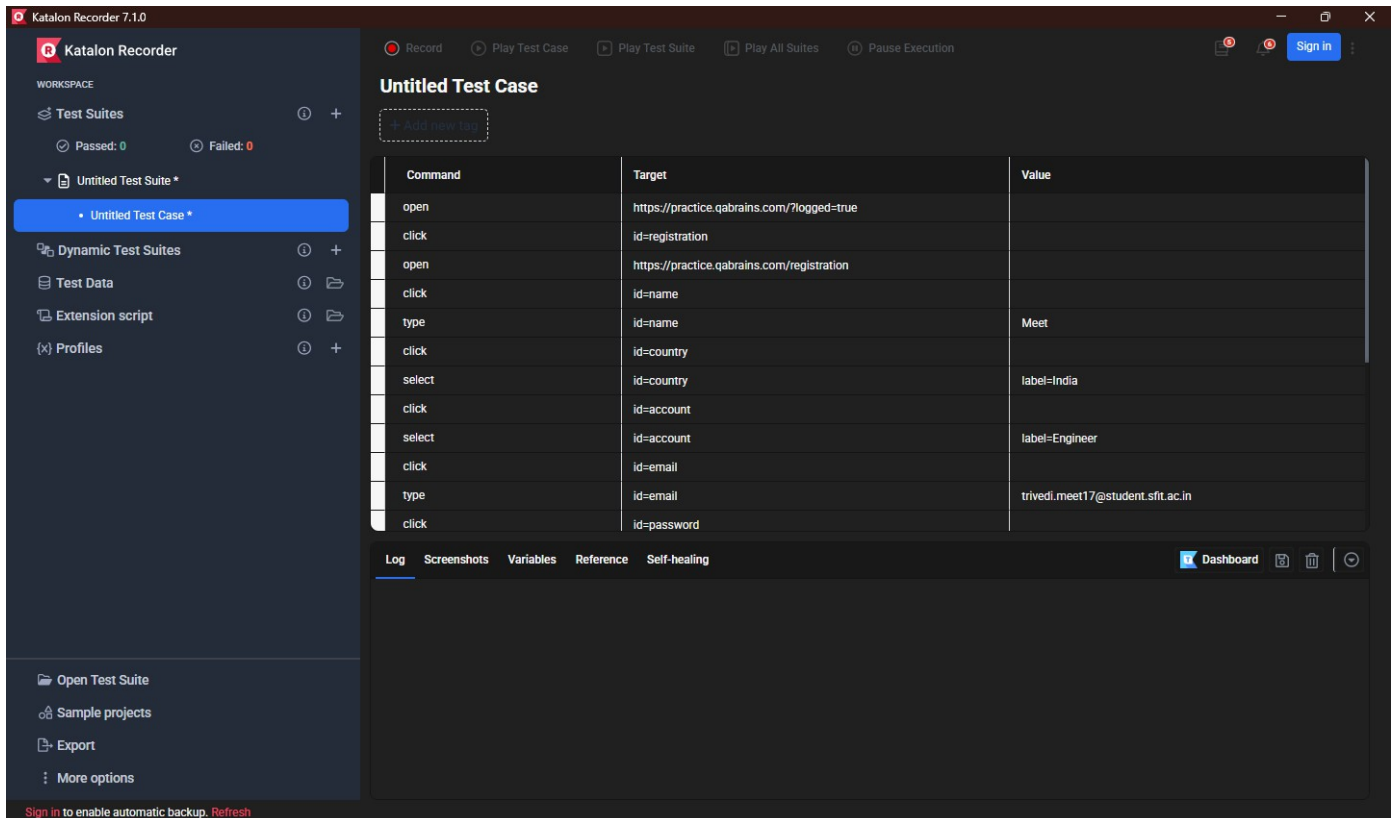


Make sure recording is on



Stop the recording and play test cases

Check Katalon Recorder



The screenshot shows the Katalon Recorder 7.1.0 interface. On the left is the 'WORKSPACE' sidebar with options like 'Test Suites', 'Dynamic Test Suites', 'Test Data', 'Extension script', and 'Profiles'. The main area is titled 'Untitled Test Case' and contains a table with three columns: 'Command', 'Target', and 'Value'.

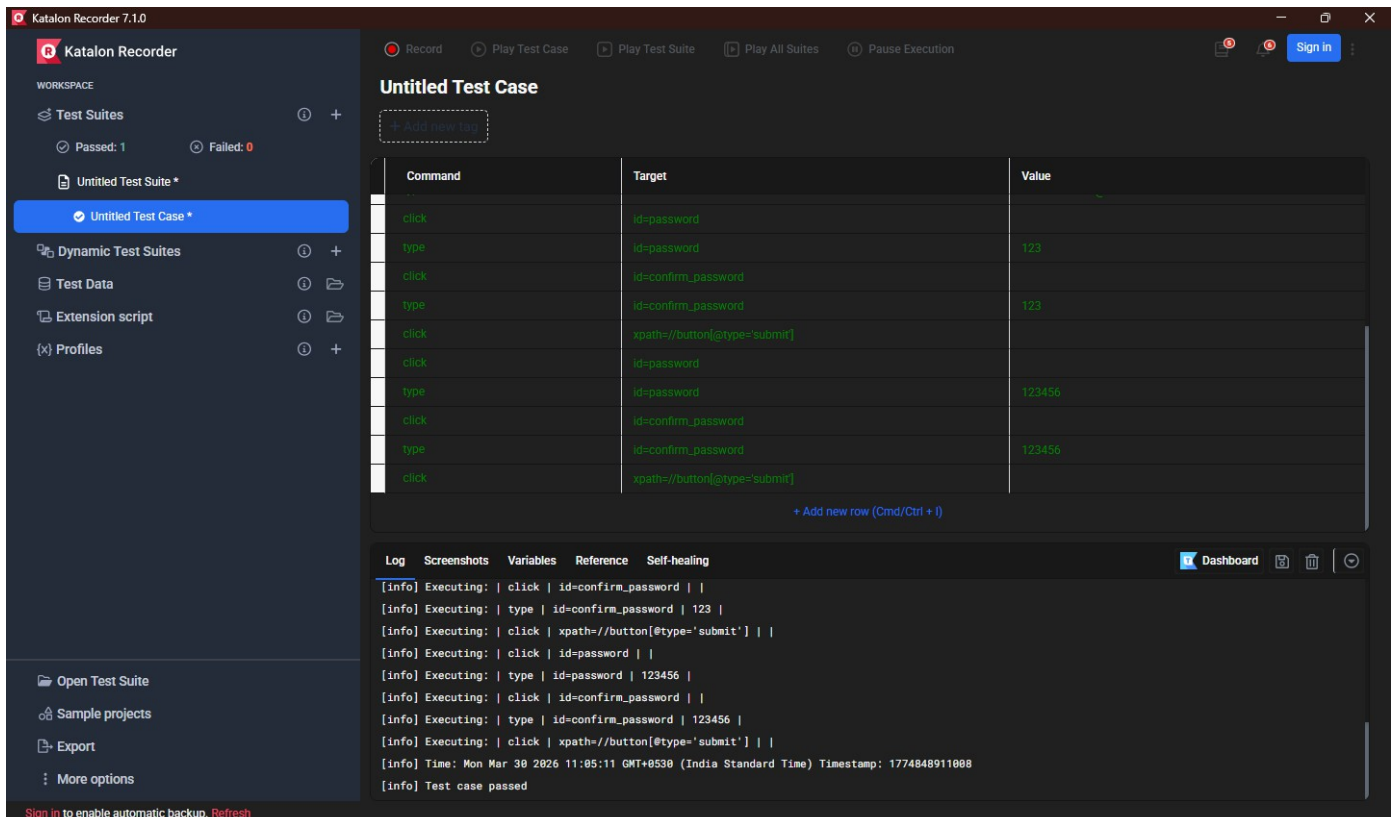
Command	Target	Value
open	https://practice.qabrain.com/?logged=true	
click	id=registration	
open	https://practice.qabrain.com/registration	
click	id=name	
type	id=name	Meet
click	id=country	
select	id=country	label=India
click	id=account	
select	id=account	label=Engineer
click	id=email	
type	id=email	trivedi.meet17@student.sft.ac.in
click	id=password	

At the bottom of the main area, there is a 'Log' tab and a 'Dashboard' button.

Click on play test cases

Check the output

Test case Passed



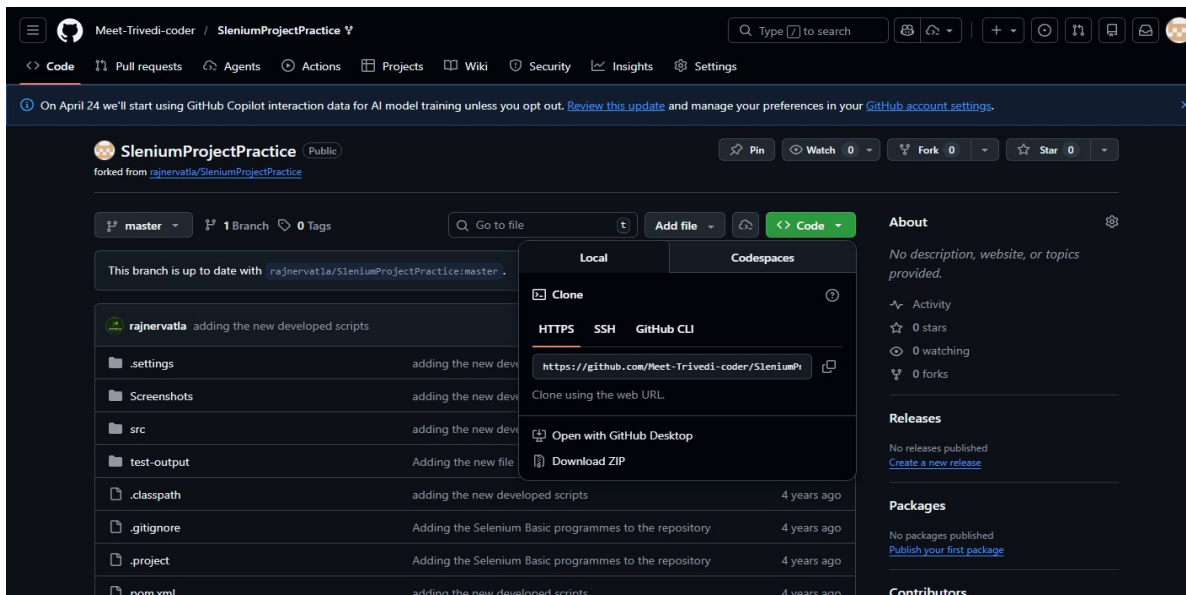
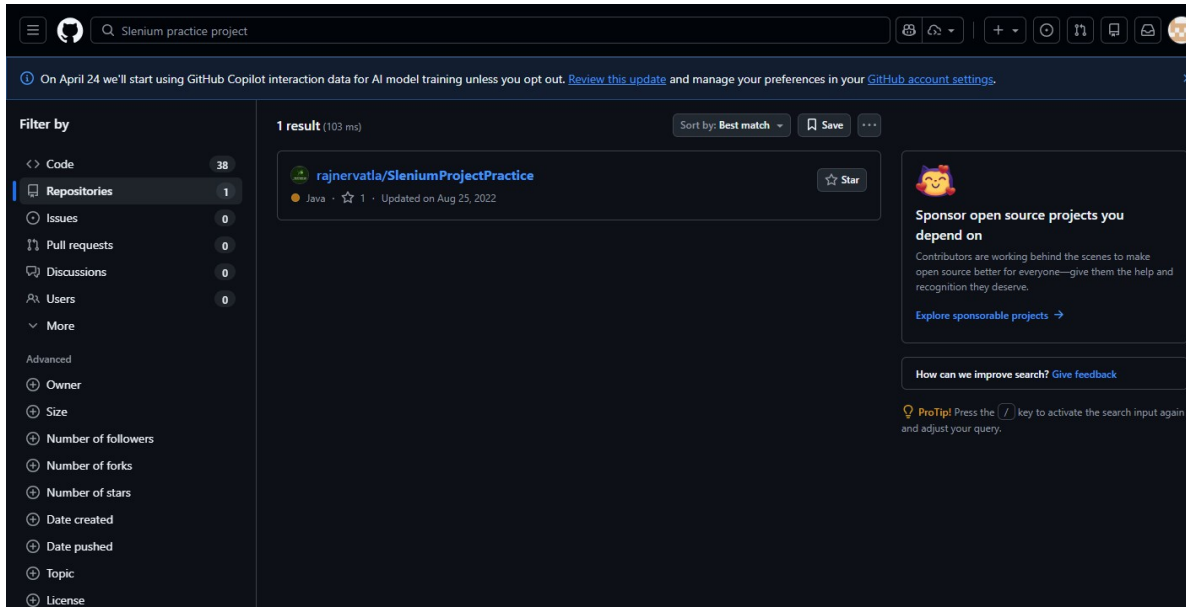
The screenshot shows the Katalon Recorder 7.1.0 interface after the test case has been executed. The 'Untitled Test Case' table is now empty, and the 'Log' tab is active, displaying the execution steps.

Command	Target	Value
---------	--------	-------

The log shows the following execution steps:

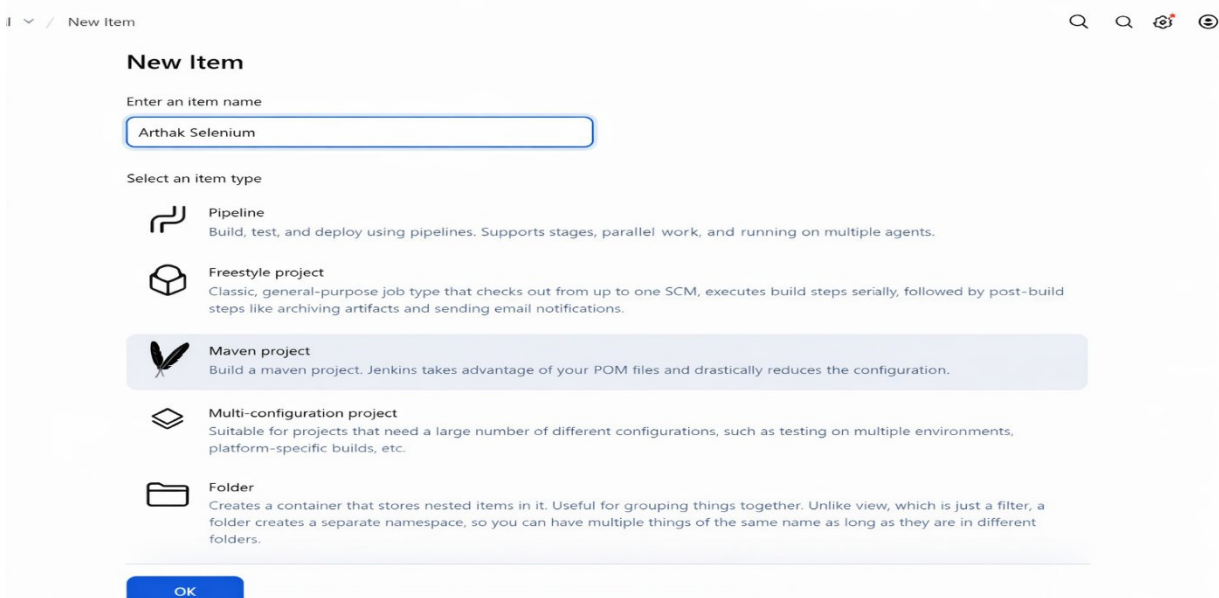
```
[info] Executing: | click | id-confirm_password | |
[info] Executing: | type | id-confirm_password | 123 |
[info] Executing: | click | xpath=//button[@type='submit'] | |
[info] Executing: | click | id=password | |
[info] Executing: | type | id=password | 123456 |
[info] Executing: | click | id-confirm_password | |
[info] Executing: | type | id-confirm_password | 123456 |
[info] Executing: | click | xpath=//button[@type='submit'] | |
[info] Time: Mon Mar 30 2026 11:05:11 GMT+0530 (India Standard Time) Timestamp: 1774846911008
[info] Test case passed
```

Go to Github and fork Slenium Project Practice repository

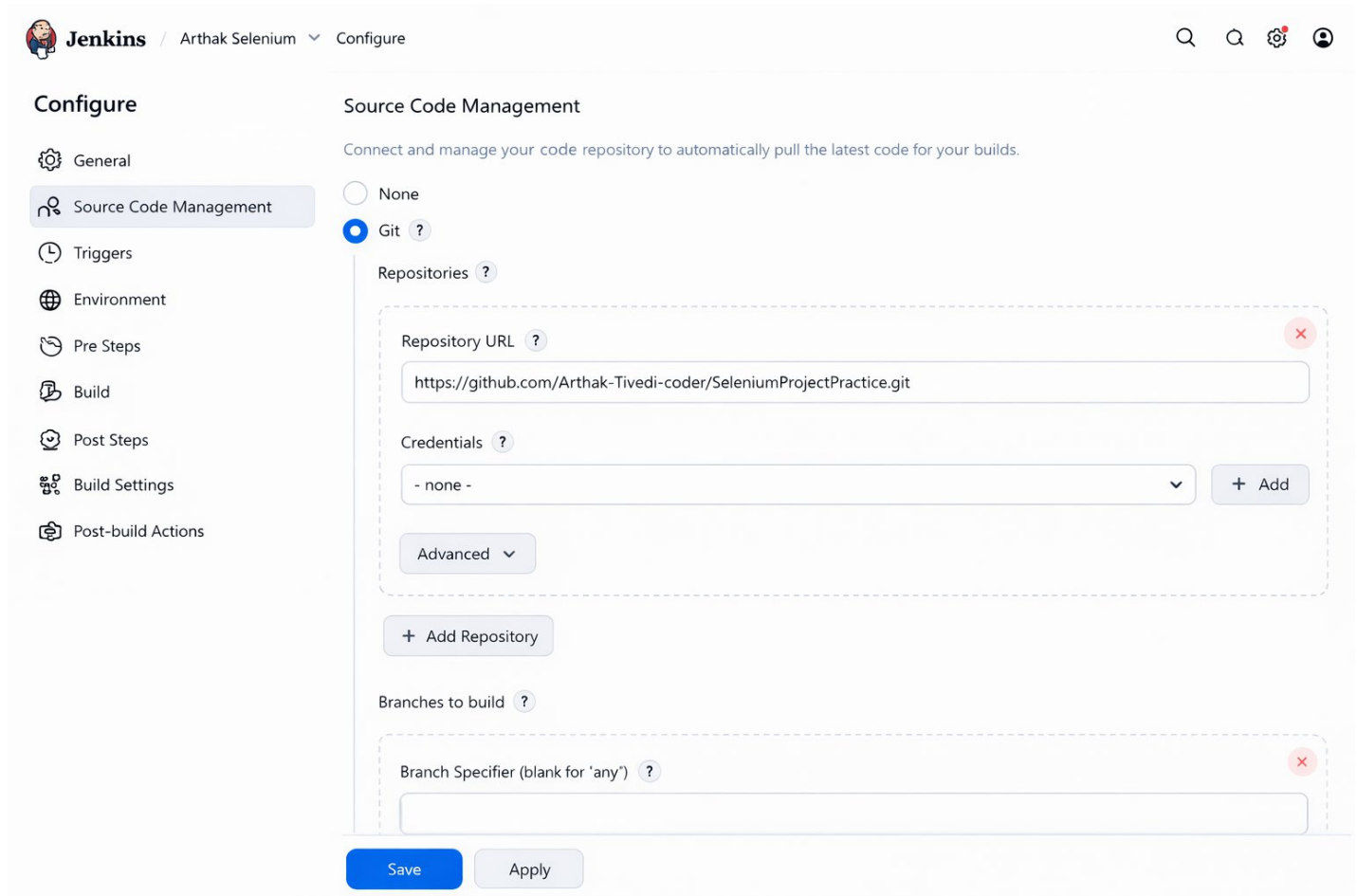


Copy URL

Go to Jenkins, New Project, Maven project



SCM : Git, Give Repository URL



The image shows the Jenkins 'Configure' page for a job named 'Arthak Selenium'. The 'Source Code Management' tab is selected in the left sidebar. The main content area is titled 'Source Code Management' and includes a description: 'Connect and manage your code repository to automatically pull the latest code for your builds.' There are two radio buttons: 'None' and 'Git'. The 'Git' option is selected. Below this, there is a 'Repositories' section with a dashed border. Inside, there is a 'Repository URL' field containing 'https://github.com/Arthak-Tivedi-coder/SeleniumProjectPractice.git', a 'Credentials' dropdown menu set to '- none -', and an 'Advanced' dropdown. Below the repository section is a '+ Add Repository' button. Further down is a 'Branches to build' section with a 'Branch Specifier (blank for 'any')' field. At the bottom of the configuration area are 'Save' and 'Apply' buttons.

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/Arthak-Tivedi-coder/SeleniumProjectPractice.git

Credentials ?

- none -

+ Add

Advanced

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

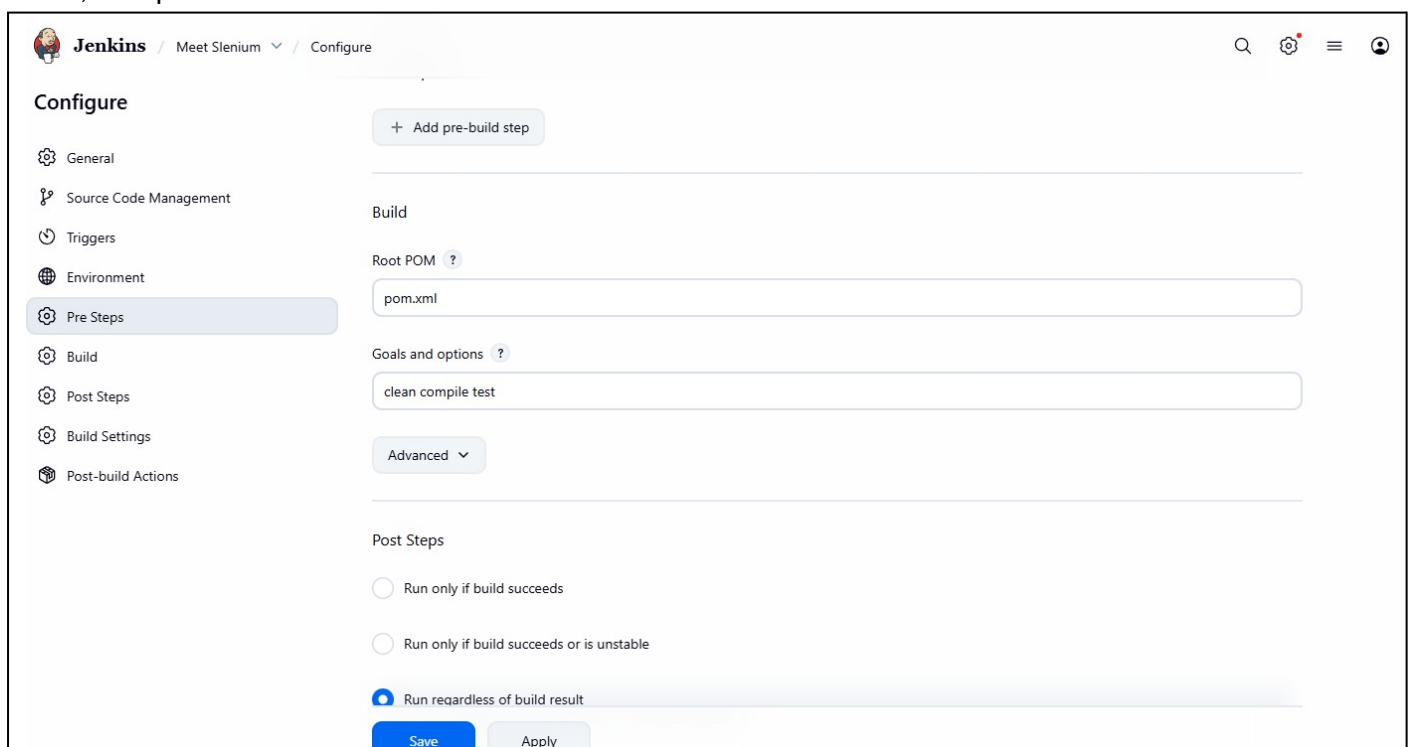
Save Apply

Build

Root POM : POM.XML

Goal and Options:

Clean, Compile and Test



The image shows the Jenkins 'Configure' page for a job named 'Meet Slenium'. The 'Build' tab is selected in the left sidebar. The main content area is titled 'Build' and includes a description: 'Build'. There is a '+ Add pre-build step' button at the top. Below this, there is a 'Root POM' field containing 'pom.xml'. Below that is a 'Goals and options' field containing 'clean compile test'. Below the goals and options section is an 'Advanced' dropdown. Further down is a 'Post Steps' section with three radio buttons: 'Run only if build succeeds', 'Run only if build succeeds or is unstable', and 'Run regardless of build result'. The 'Run regardless of build result' option is selected. At the bottom of the configuration area are 'Save' and 'Apply' buttons.

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build**
- Post Steps
- Build Settings
- Post-build Actions

+ Add pre-build step

Build

Root POM ?

pom.xml

Goals and options ?

clean compile test

Advanced

Post Steps

☐ Run only if build succeeds

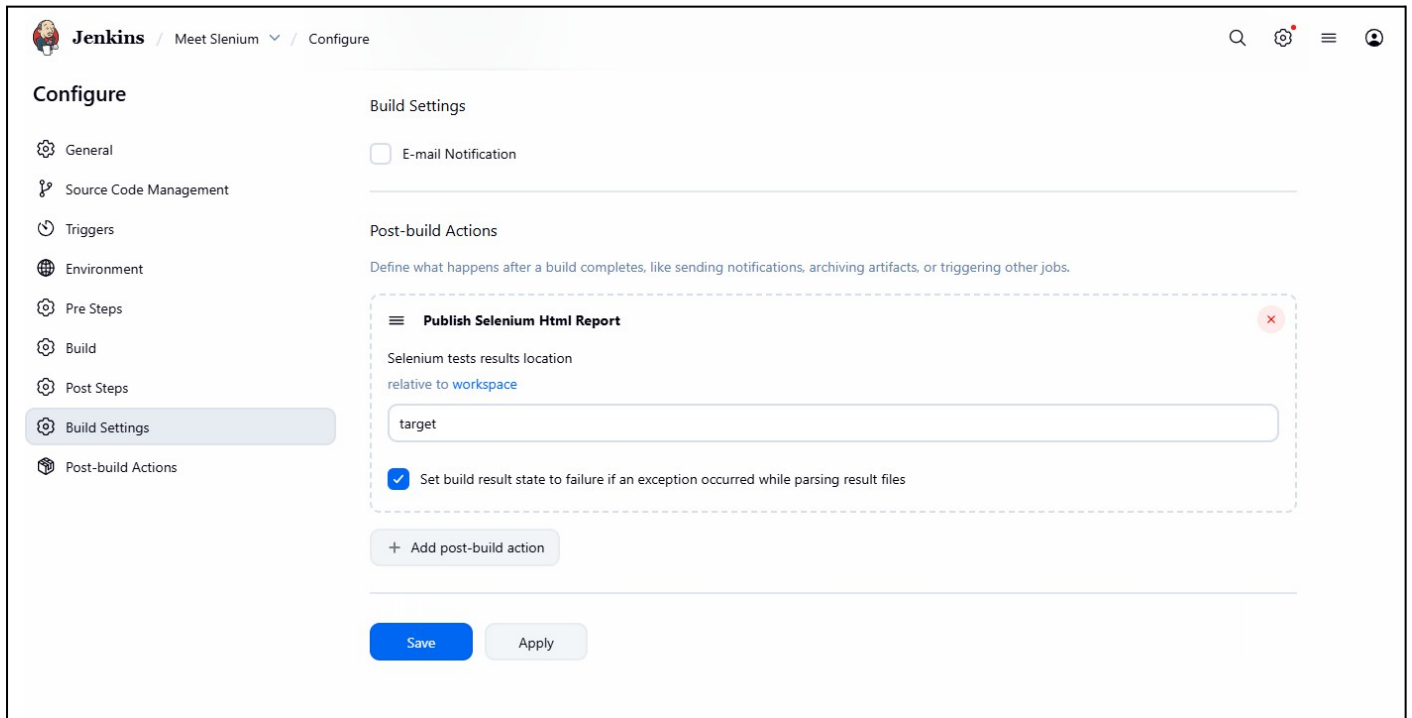
☐ Run only if build succeeds or is unstable

☒ Run regardless of build result

Save Apply

Post Build Action : Publish Selenium HTML Report

Apply and Save



Jenkins / Meet Slenium / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings**
- Post-build Actions

Build Settings

☐ E-mail Notification

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Publish Selenium Html Report

Selenium tests results location
relative to workspace

target

☒ Set build result state to failure if an exception occurred while parsing result files

+ Add post-build action

Save **Apply**

Build nOw

Check Console output

And selenium HTML Report



Jenkins / Meet Slenium / #2

```
unless at least one processor is specified by name (-processor), or a search
path is specified (--processor-path, --processor-module-path), or annotation
processing is enabled explicitly (-proc:only, -proc:full).
Use -Xlint:options to suppress this message.
Use -proc:none to disable annotation processing.
[INFO] /C:/ProgramData/Jenkins/.jenkins/workspace/Meet Slenium/src/test/java/widgets/ProgressBar.java: Some input files use or
override a deprecated API.
[INFO] /C:/ProgramData/Jenkins/.jenkins/workspace/Meet Slenium/src/test/java/widgets/ProgressBar.java: Recompile with -
Xlint:deprecation for details.
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ Selenium ---
[JENKINS] Recording test results
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.412 s
[INFO] Finished at: 2026-03-30T11:31:13+05:30
[INFO] -----
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\Meet Slenium\pom.xml to com.qa/Selenium/0.0.1-SNAPSHOT/Selenium-
0.0.1-SNAPSHOT.pom
channel stopped
Publishing Selenium report...
Copying the reports.
Finished: SUCCESS
```

Jenkins 2.533